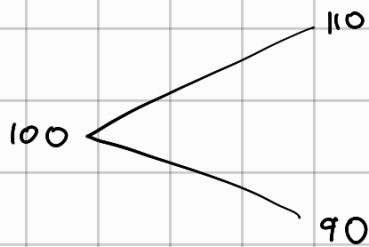


3a. Assume the following.

B is cash in portfolio.

$$\Delta S_u + B = 10$$

$$\Delta S_d + B = 0$$



$$\Delta S_u + B - 10 = \Delta S_d + B$$

$$110\Delta - 10 = \Delta 90$$

$$20\Delta = 10$$

$$\Delta = 0.5$$

$$0.5(110) + B = 10$$

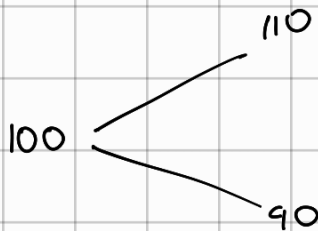
$$55 + B = 10$$

$$B = -45$$

$$\therefore 0.5(100) + (-45) = S$$

b. $\Delta S_u + B e^{rT} = 10$

$$\Delta S_d + B e^{rT} = 0$$



$$\Delta 110 + \cancel{B e^{0.05}} - 10 = \Delta 90 + \cancel{B e^{0.05}}$$

$$\Delta = 0.5$$

$$0.05 \times 110 + B e^{0.05} = 10$$

$$B e^{0.05} = -45$$

$$B = -42.81$$

$$0.5(100) - 42.81 = 7.19$$

S. PDE

$$dx_t = \sigma dw_t$$

$$dv_t = \frac{\partial V}{\partial x} dx_t + \frac{\partial^2 V}{\partial x^2} (dx_t)^2$$

as $V(x, t)$ is also a function of x

$$V(x, t) = \frac{V_t}{B_t}$$

$$\therefore \frac{\partial V}{\partial B_t} = -\frac{V_t}{B_t^2} dB_t$$

$$\frac{\partial V}{\partial \hat{V}} = -\frac{V_t}{B_t}$$

$$dV_t = \frac{\partial V}{\partial B_t} dB_t + \frac{\partial V}{\partial \hat{V}} dV_t$$

$$= -\frac{\hat{V}_t}{B_t} r B_t dt + \frac{1}{2} \frac{1}{B_t} [dV_t]$$

$$= -r V_t dt + \frac{1}{B_t} \left[\frac{\partial V_t}{\partial x} dx_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2} (dx_t)^2 \right]$$

$$= -r V_t dt + \frac{1}{B_t} \left[\frac{\partial V}{\partial x} \sigma dw_t + \frac{\partial^2 V}{\partial x^2} \frac{\sigma^2}{2} dt \right]$$

$$= \frac{1}{B_t} \left(\frac{\sigma^2}{2} \frac{\partial^2 V}{\partial x^2} - r V_t \right) dt + \frac{1}{B_t} \left(\frac{\partial V}{\partial x} \sigma dw_t \right)$$

$$\therefore \text{PDE} : \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial x^2} - r V_t = 0$$